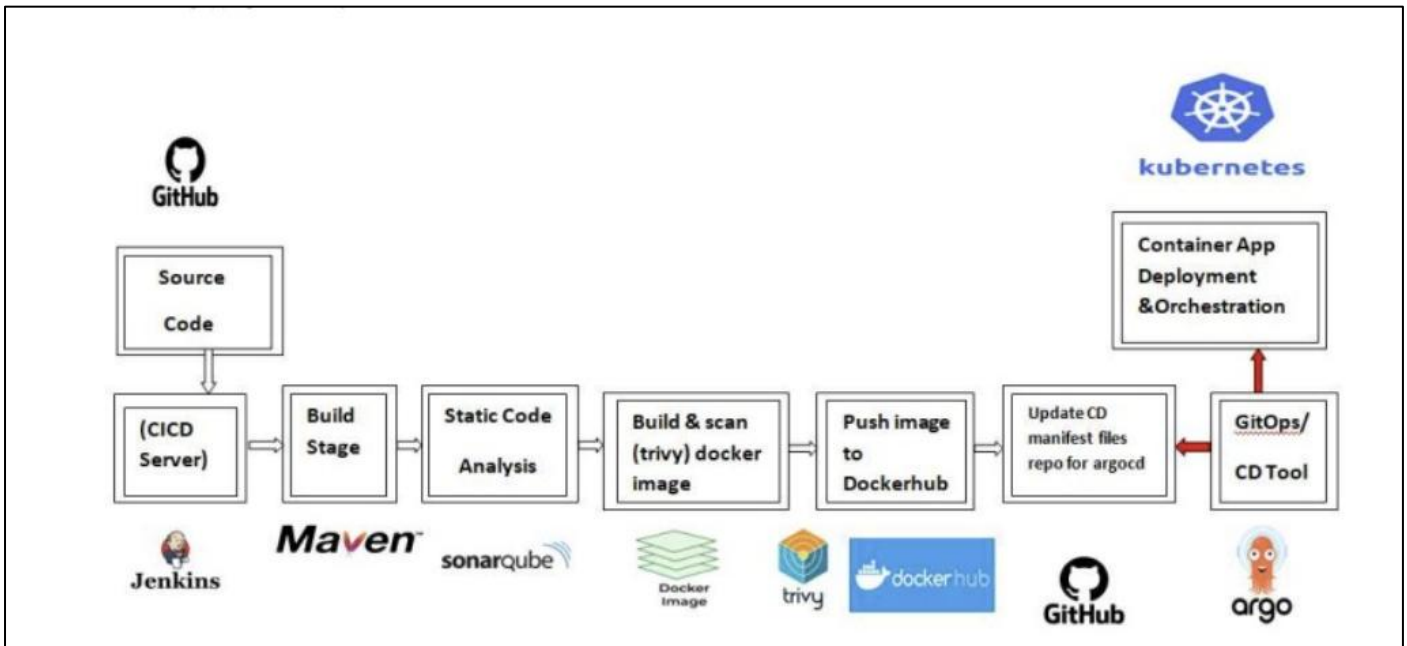
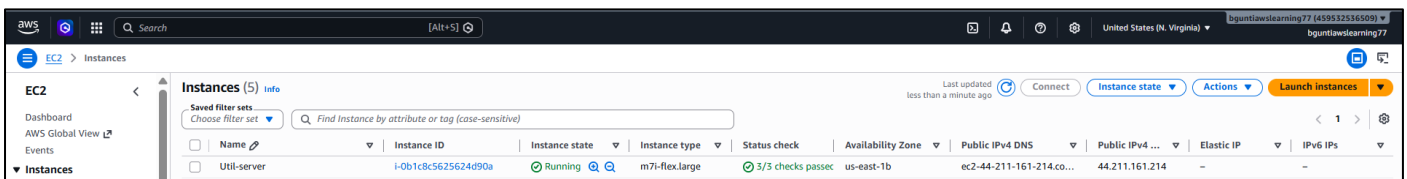


END-TO-END Project – CI/CD Project using ARGO CD – GITOPS Model



Launch Ec2 instance – Util server



1. Install Jenkins

```
sudo yum update -y
```

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \
```

```
https://pkg.jenkins.io/rpm-stable/jenkins.repo
```

```
sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins.io-2026.key
```

```
sudo yum upgrade -y
```

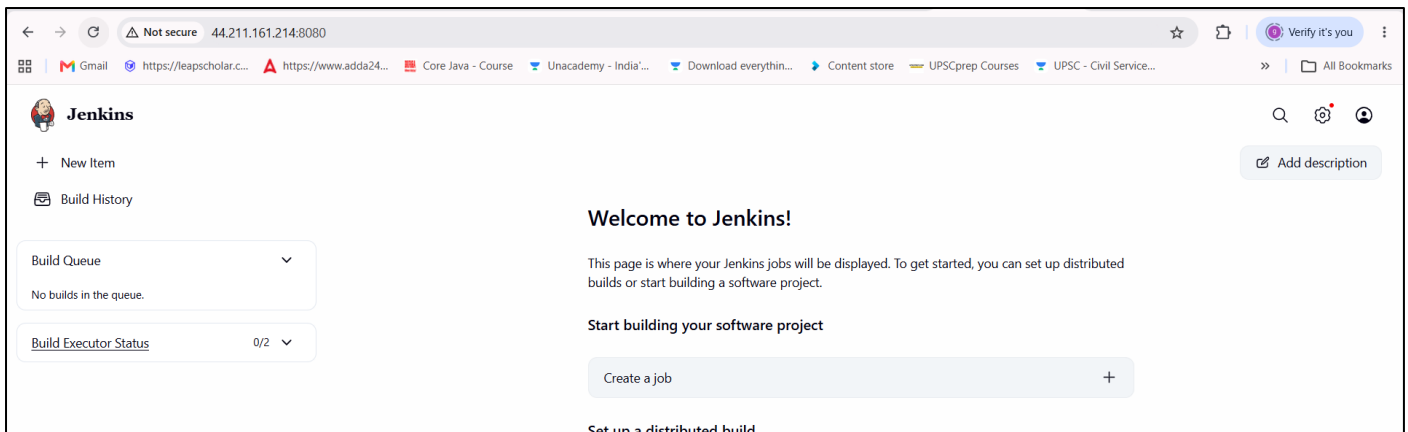
```
sudo yum install java-21-amazon-corretto -y
```

```
sudo yum install jenkins -y
```

```
sudo systemctl enable jenkins
```

```
sudo systemctl start jenkins
```

```
sudo systemctl status jenkins
```



2. Install Docker and Configure – to generate Docker image

```
[root@ip-172-31-90-81 ~]# yum install docker -y
```

```
[root@utilserver ~]# yum install git -y
```

```
[root@ip-172-31-90-81 ~]# docker images
```

Cannot connect to the Docker daemon at unix:///var/run/docker.sock. Is the docker daemon running?

```
[root@ip-172-31-90-81 ~]# systemctl start docker
```

```
[root@ip-172-31-90-81 ~]# docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
------------	-----	----------	---------	------

```
[root@ip-172-31-90-81 ~]# exit
```

logout

```
[ec2-user@ip-172-31-90-81 ~]$ docker images
```

permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.sock: connect: permission denied

```
[ec2-user@ip-172-31-90-81 ~]$ sudo su -
```

Last login: Sat Jun 13 15:41:53 UTC 2026 on pts/1

```
[root@utilserver ~]# cat /etc/group
```

ec2-user:x:1000:

jenkins:x:992:

docker:x:991:

add ec2 user , jenkins user to docker group

```
[root@utilserver ~]# usermod -aG docker jenkins
```

```
[root@utilserver ~]# usermod -aG docker ec2-user
```

```
[root@utilserver ~]# cat /etc/group
```

```
ec2-user:x:1000:
```

```
jenkins:x:992:
```

```
docker:x:991:jenkins,ec2-user
```

```
[root@utilserver ~]# systemctl restart docker
```

```
[root@utilserver ~]# sudo chmod 666 /var/run/docker.sock
```

```
[root@utilserver ~]# exit
```

logout

```
[ec2-user@ip-172-31-90-81 ~]$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
------------	-----	----------	---------	------

```
[ec2-user@ip-172-31-90-81 ~]$
```

Now ec2-user , jenkins user can execute docker commands

3. AWS Cli will be auto installed in EC2, you need to configure it

```
[ec2-user@utilserver ~]$ sudo su -
```

```
[root@utilserver ~]# aws --version
```

```
aws-cli/2.33.15 Python/3.9.25 Linux/6.1.174-217.345.amzn2023.x86_64  
source/x86_64.amzn.2023
```

```
[root@utilserver ~]# aws sts get-caller-identity
```

Unable to locate credentials. You can configure credentials by running "aws login".

```
[root@utilserver ~]# aws configure
```

```
AWS Access Key ID [None]: AKIAWV7SGBK6VZL2VURO
```

```
AWS Secret Access Key [None]: GJd7JbmUf8jies0ebKV4sEQIYx8PRg4HIHsvcyL
```

```
Default region name [None]: us-east-1
```

```
[root@utilserver ~]# aws sts get-caller-identity
```

```
{  
  "UserId": "459532536509",  
  "Account": "459532536509",  
  "Arn": "arn:aws:iam::459532536509:root"  
}
```

4. Kubectl Installation commands

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s  
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s  
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl.sha256"
```

```
echo "$(cat kubectl.sha256) kubectl" | sha256sum --check
```

```
sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
```

```
kubectl version --client
```

```
[root@utilserver ~]# curl -LO "https://dl.k8s.io/release/$(curl -L -s  
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s  
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl.sha256"
```

```
echo "$(cat kubectl.sha256) kubectl" | sha256sum --check
```

```
sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
```

```
kubectl version --client
```

5. EKCTL Installation

for ARM systems, set ARCH to: `arm64`, `armv6` or `armv7`

ARCH=amd64

PLATFORM=\$(uname -s)_\$ARCH

curl -sLO [https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_\\${PLATFORM}.tar.gz](https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_${PLATFORM}.tar.gz)

tar -xzf eksctl_\${PLATFORM}.tar.gz -C /tmp && rm -rf eksctl_\${PLATFORM}.tar.gz

sudo install -m 0755 /tmp/eksctl /usr/local/bin && rm /tmp/eksctl

[root@utilserver ~]# ARCH=amd64

PLATFORM=\$(uname -s)_\$ARCH

curl -sLO [https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_\\${PLATFORM}.tar.gz](https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_${PLATFORM}.tar.gz)

tar -xzf eksctl_\${PLATFORM}.tar.gz -C /tmp && rm eksctl_\${PLATFORM}.tar.gz

sudo install -m 0755 /tmp/eksctl /usr/local/bin && rm /tmp/eksctl

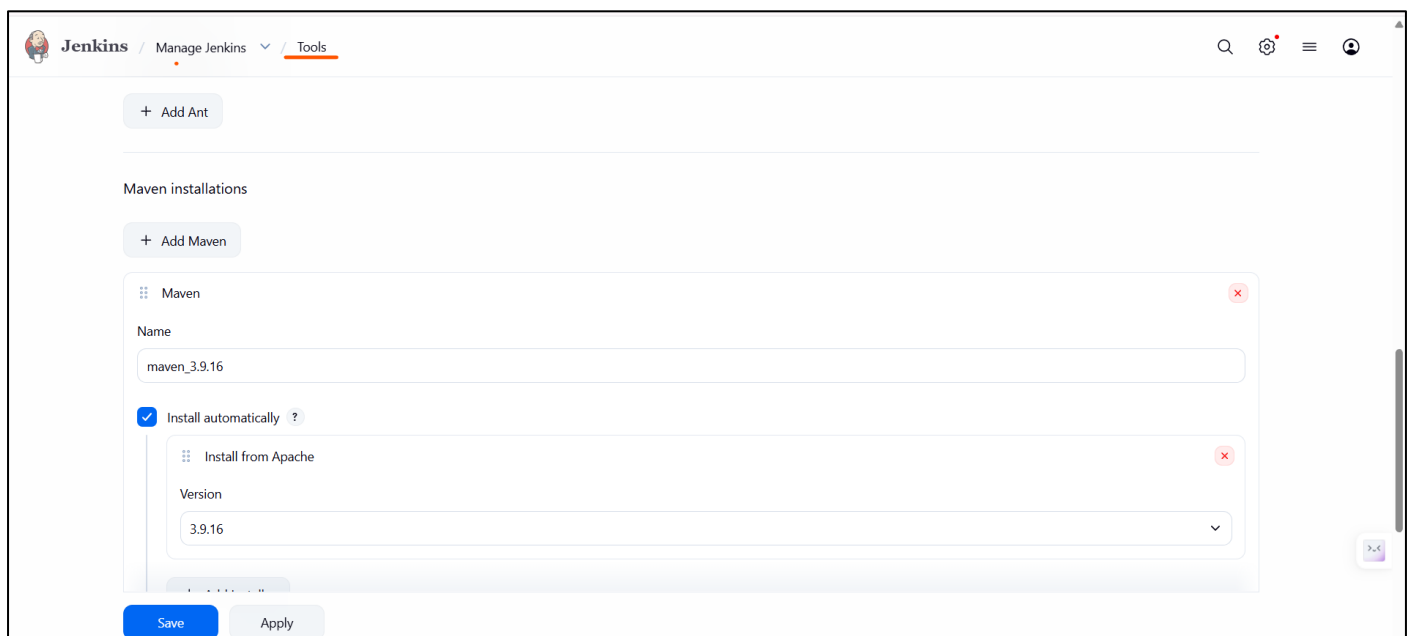
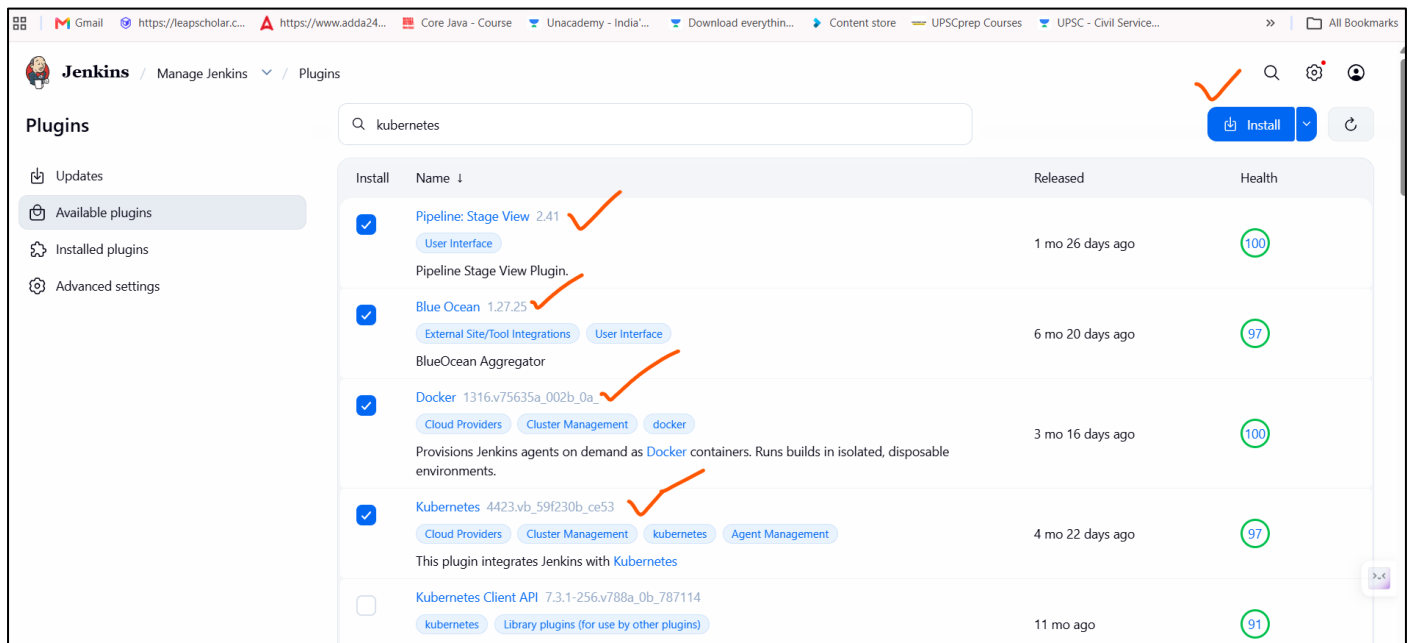
rm: remove regular file 'eksctl_Linux_amd64.tar.gz'? y

rm: remove regular file '/tmp/eksctl'? y

[root@utilserver ~]# eksctl version

0.227.0

6. Setup jenkins



7. Install Sonar Qube container

```
[root@utilserver ~]# docker run -itd --name sonarcontainer -p 9000:9000 sonarqube
```

Unable to find image 'sonarqube:latest' locally

latest: Pulling from library/sonarqube

cb259a83ac3d: Pull complete

d264b91a55d8: Pull complete

28e0126b19f1: Pull complete

4f4fb700ef54: Pull complete

72c6a52a9f2d: Pull complete

d385704da112: Pull complete

ba5c348e0e55: Pull complete

Digest:

sha256:4671314254b0892186ebf46350e8a3a99e2505212bb6fa68bc4dc71ef3ac9c66

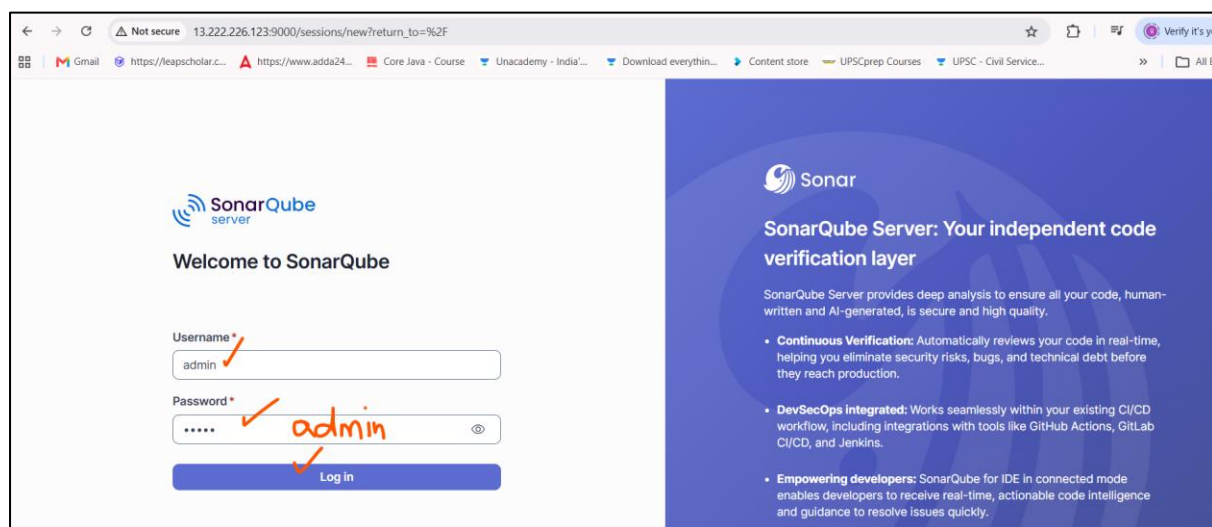
Status: Downloaded newer image for sonarqube:latest

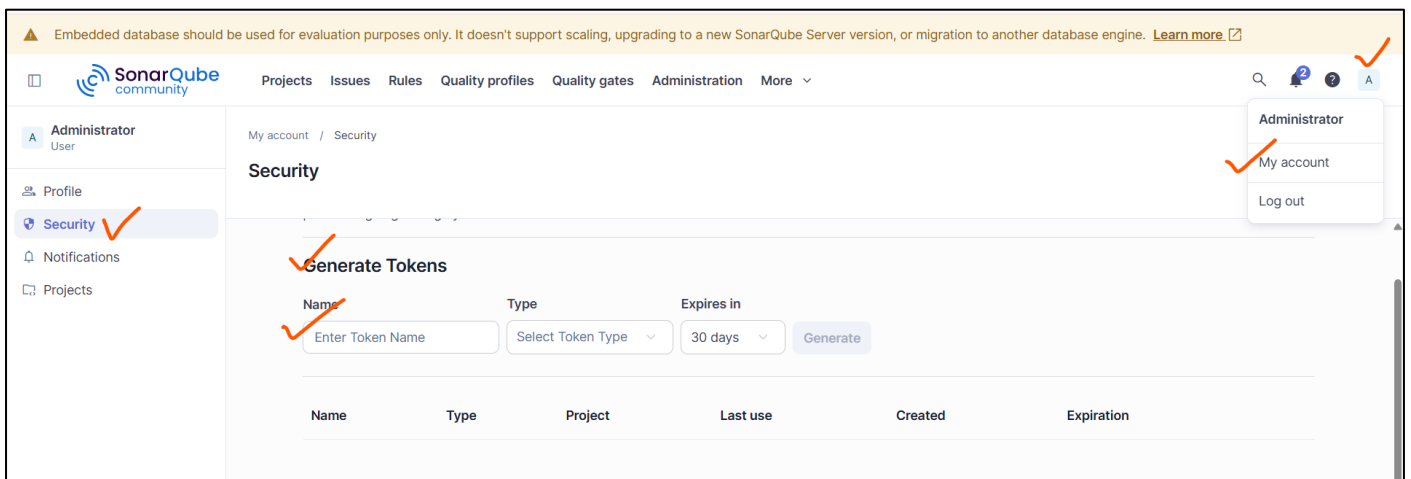
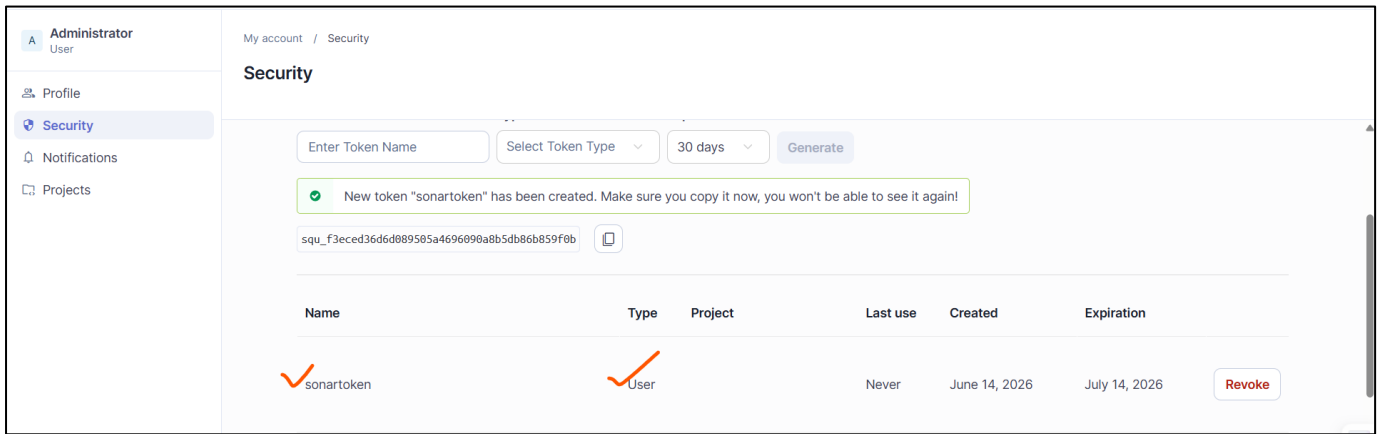
22b15dfc78944b8573b31d2ae4c4a8890ade810431eea543b3e591951f4a4504

[root@utilserver ~]# docker ps -a

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
--------------	-------	---------	---------	--------	-------

22b15dfc7894	sonarqube	"/opt/sonarqube/dock..."	16 seconds ago	Up 9 seconds	0.0.0.0:9000->9000/tcp, :::9000->9000/tcp
--------------	-----------	--------------------------	----------------	--------------	---





squ_f3eced36d6d089505a4696090a8b5db86b859f0b

8. Install Trivy

rpm -ivh xyz.rpm – to Install

rpm -Uvh xyz.rpm – To Update

rpm -ivh

https://github.com/aquasecurity/trivy/releases/download/v0.70.0/trivy_0.70.0_Linux-64bit.rpm

[root@utilserver ~]# rpm -ivh

https://github.com/aquasecurity/trivy/releases/download/v0.70.0/trivy_0.70.0_Linux-64bit.rpm

Retrieving

https://github.com/aquasecurity/trivy/releases/download/v0.70.0/trivy_0.70.0_Linux-64bit.rpm

warning: /var/tmp/rpm-tmp.qNKlvW: Header V4 RSA/SHA256 Signature, key ID 4fd9ca9f: NOKEY

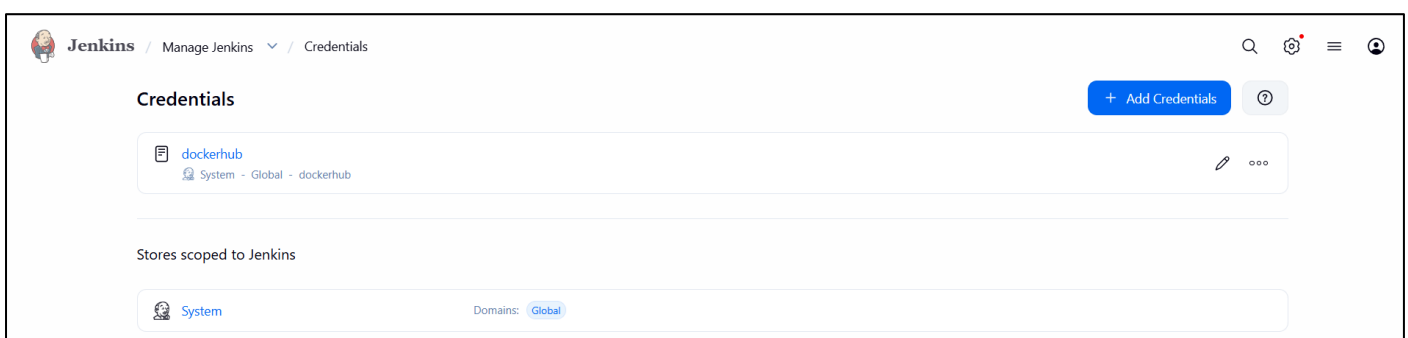
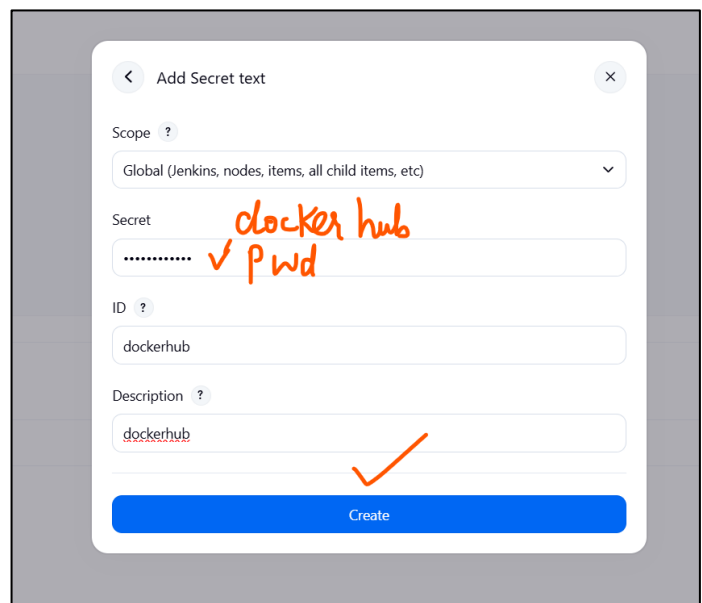
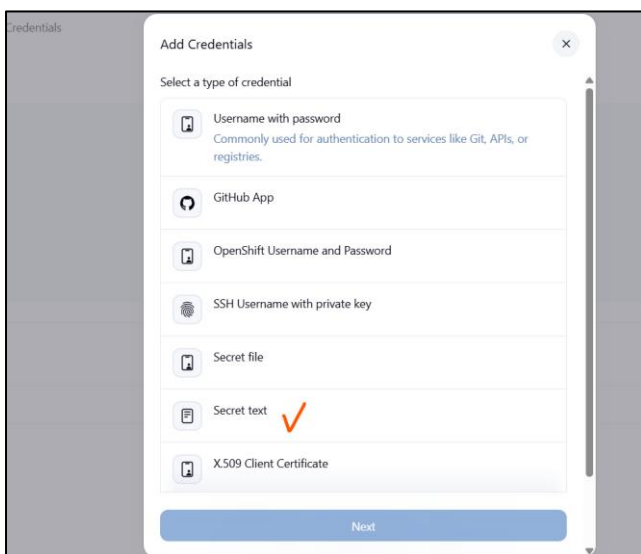
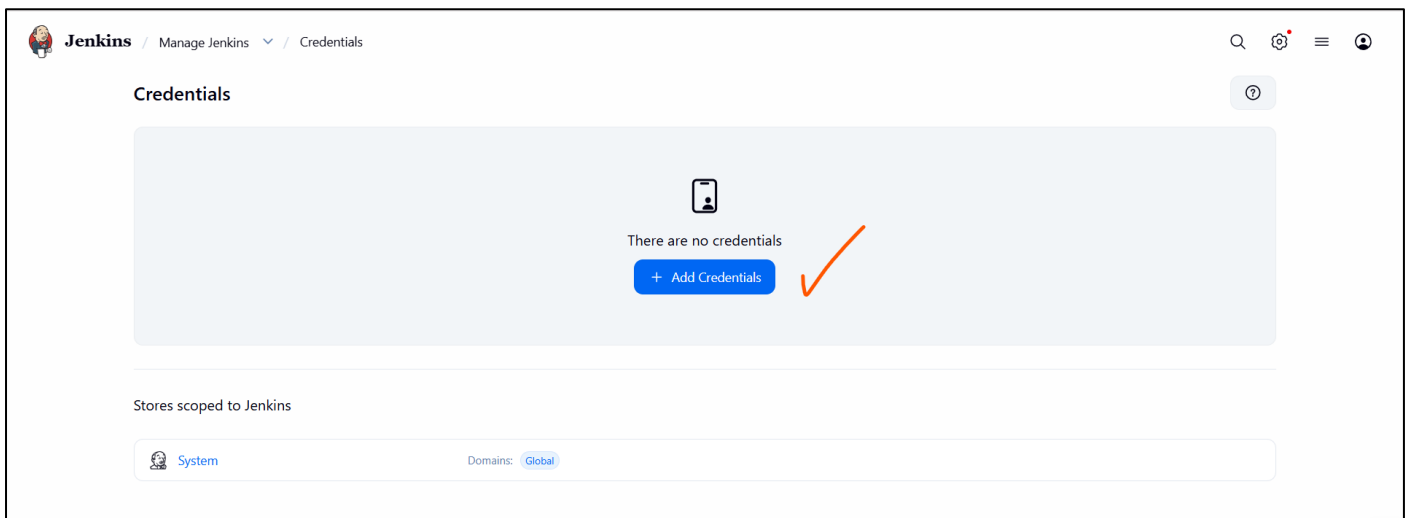
Verifying... ##### [100%]

Preparing... ##### [100%]

Updating / installing...

1:trivy-0.70.0-1

9. Add Docker Hub credentions in Jenkins



Jenkins / project-pipeline / Pipeline Syntax

Snippet Generator

Overview

This **Snippet Generator** will help you learn the Pipeline Script code which can be used to define various steps. Pick a step you are interested in from the list, configure it, click **Generate Pipeline Script**, and you will see a Pipeline Script statement that would call the step with that configuration. You may copy and paste the whole statement into your script, or pick up just the options you care about. (Most parameters are optional and can be omitted in your script, leaving them at default values.)

Steps

Sample Step

withCredentials: Bind credentials to variables

withCredentials ?

Secret values are masked on a best-effort basis to prevent *accidental* disclosure. Multiline secrets, such as the contents of a SSH private key file, are not masked. See the inline help for details and usage guidelines.

Bindings

+ Add

Generate Pipeline Script

Jenkins / project-pipeline / Pipeline Syntax

IntelliJ IDEA GDSE

withCredentials ?

Secret values are masked on a best-effort basis to prevent *accidental* disclosure. Multiline secrets, such as the contents of a SSH private key file, are not masked. See the inline help for details and usage guidelines.

Bindings

Secret text ?

Variable ?

dockerhubvar

Credentials ?

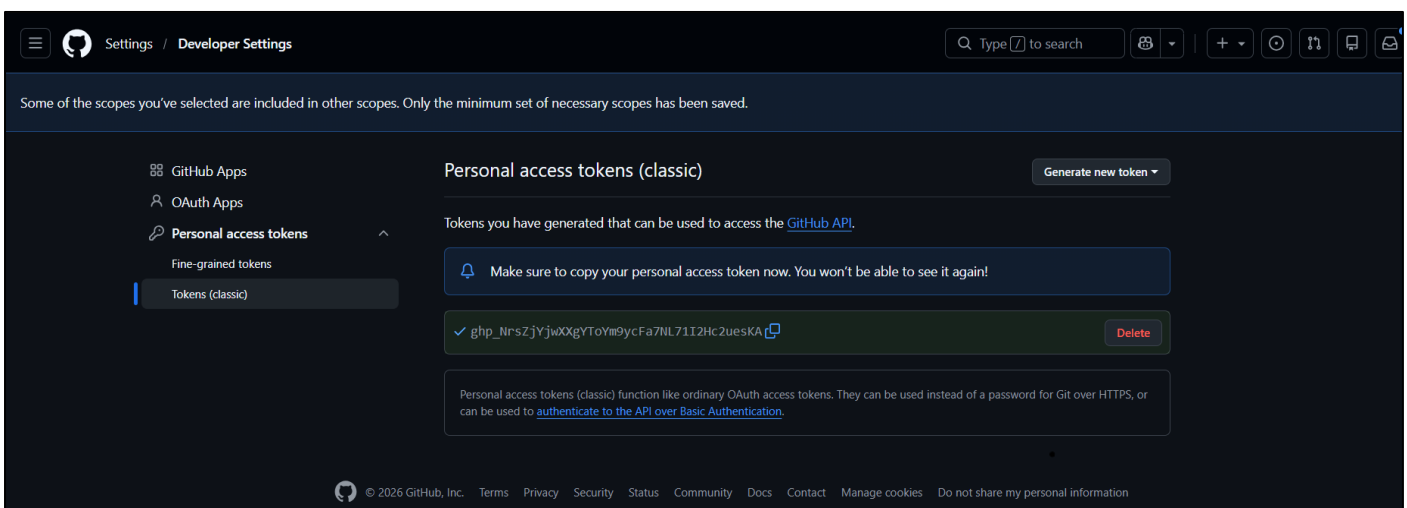
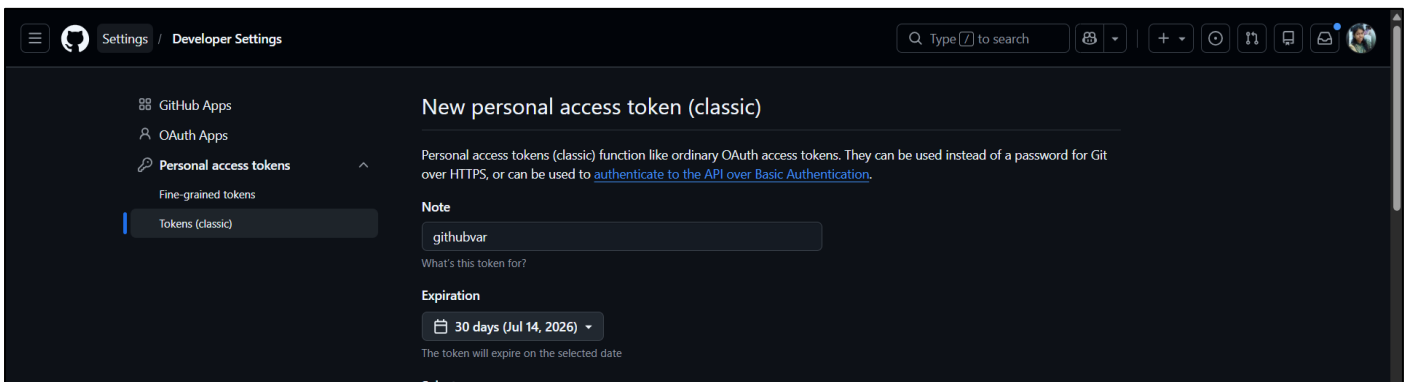
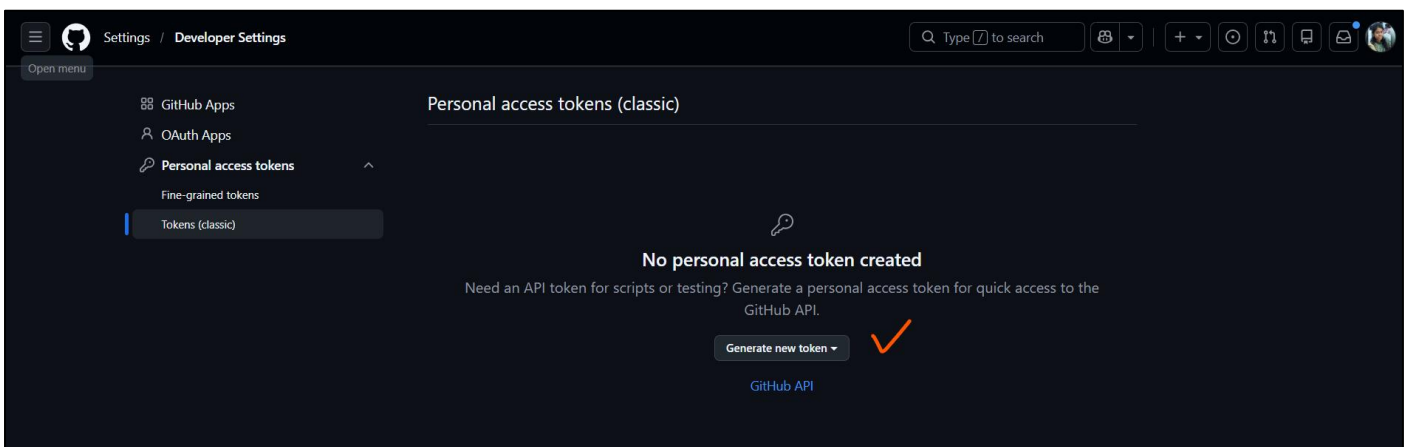
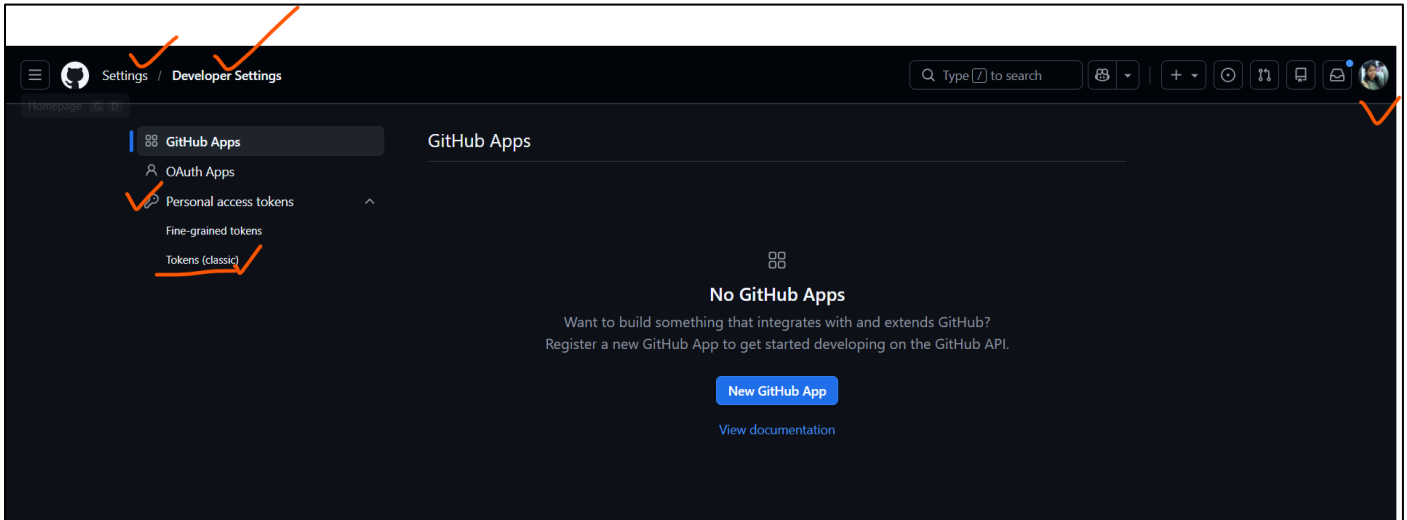
dockerhub

+ Add

Generate Pipeline Script

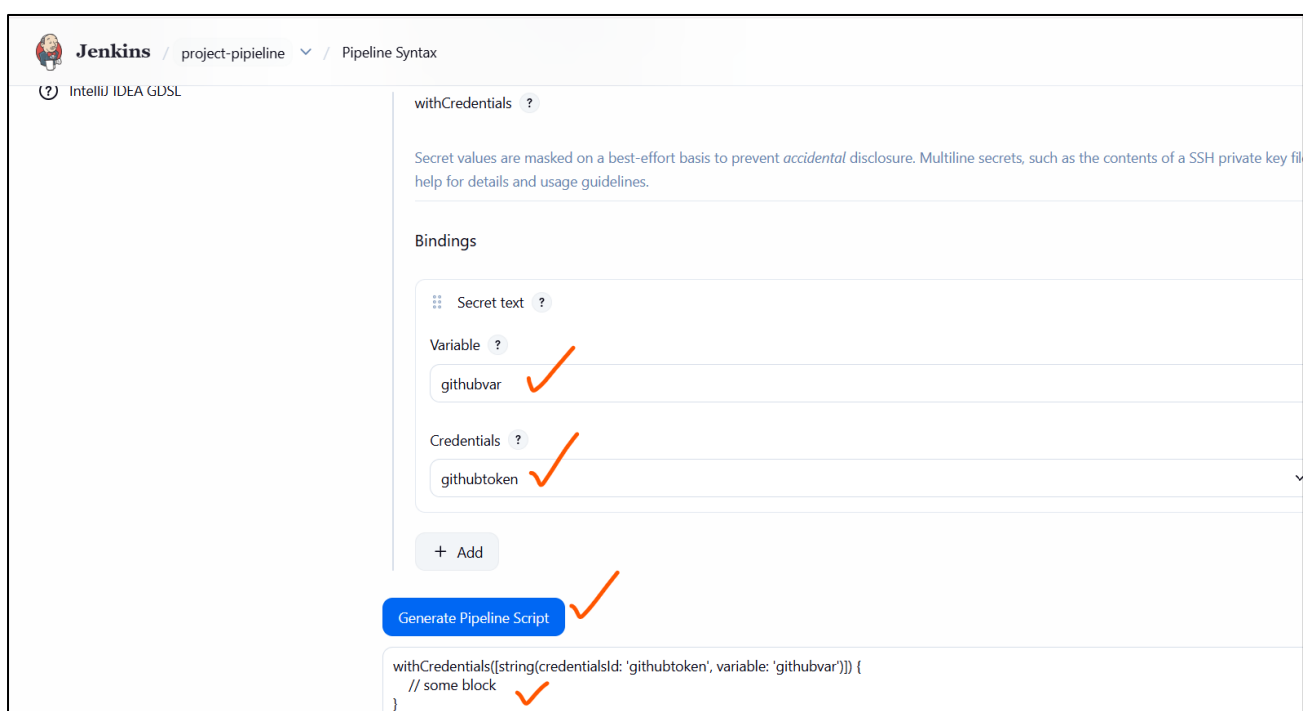
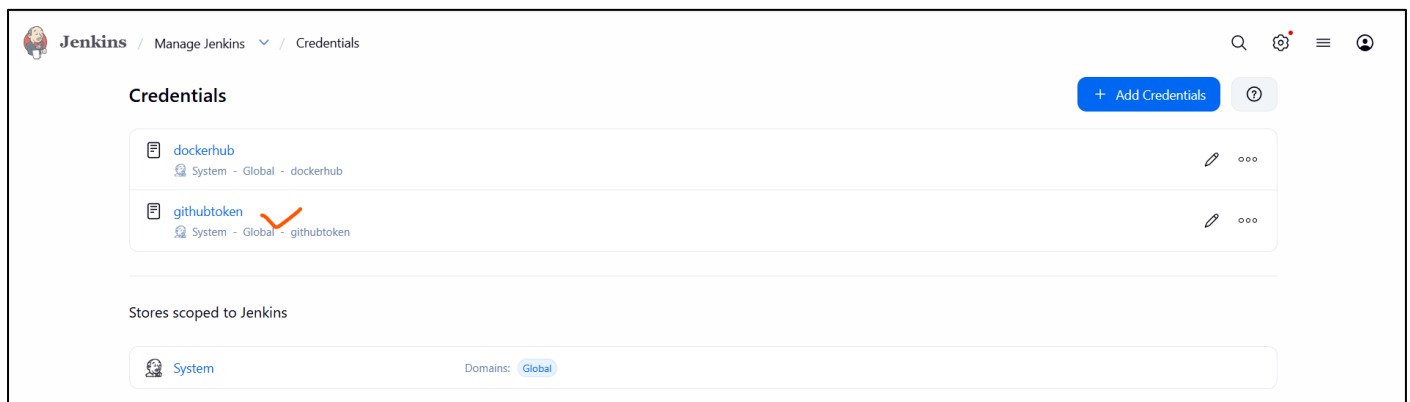
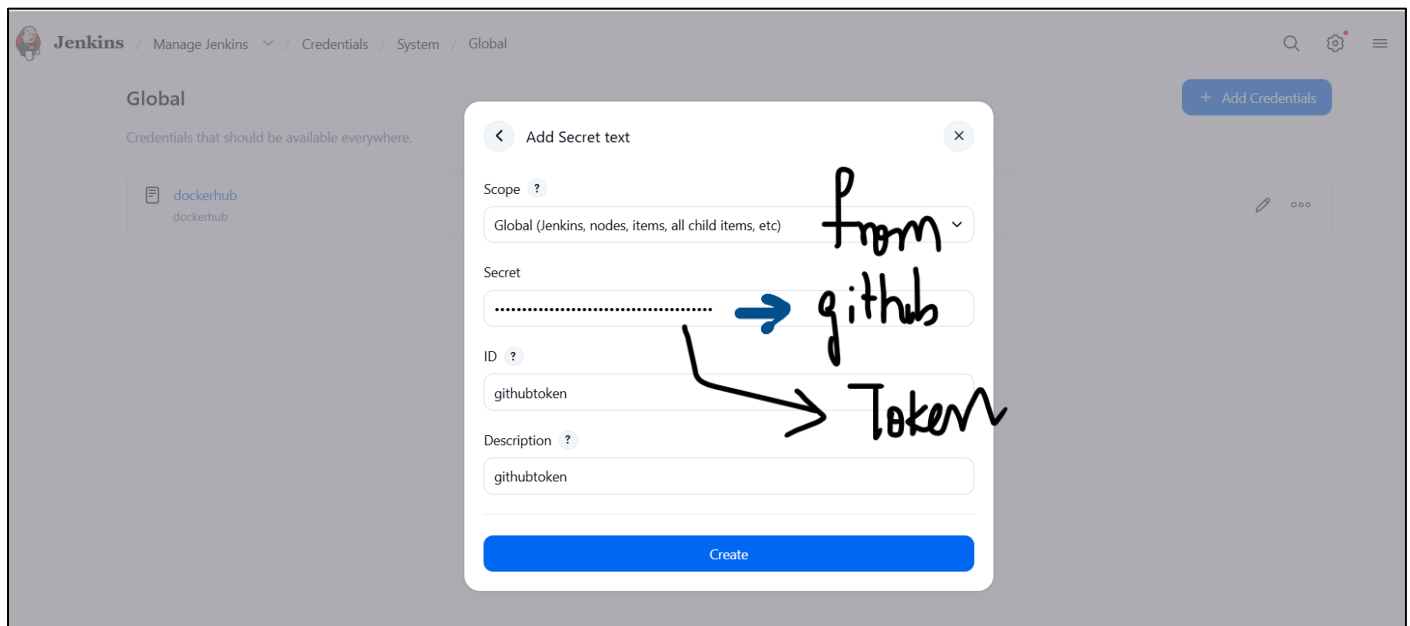
withCredentials([string(credentialsId: 'dockerhub', variable: 'dockerhubvar')]) {
// some block

10. Configure github token in git account → useful while updating the deploy.yaml

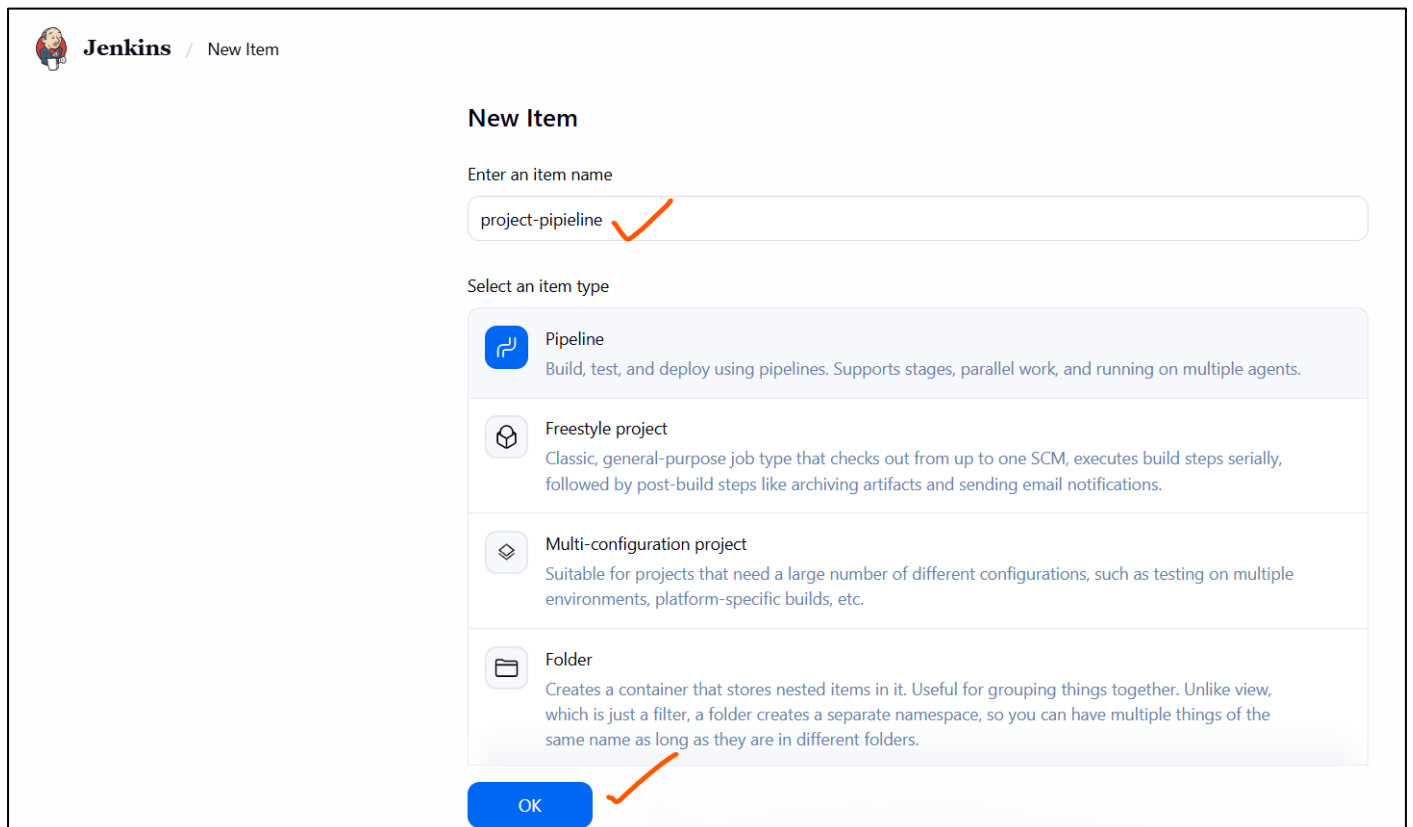


Github token

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



11.Create Jenkins CI Pipeline



The image shows the Jenkins 'New Item' configuration page. At the top, the Jenkins logo and 'New Item' are visible. Below, there's a section 'New Item' with a text input field 'Enter an item name' containing 'project-pipeline'. A dropdown menu 'Select an item type' is open, showing four options: 'Pipeline' (selected), 'Freestyle project', 'Multi-configuration project', and 'Folder'. The 'Pipeline' option is highlighted with a blue background. At the bottom, there is an 'OK' button.

Jenkins / New Item

New Item

Enter an item name

project-pipeline

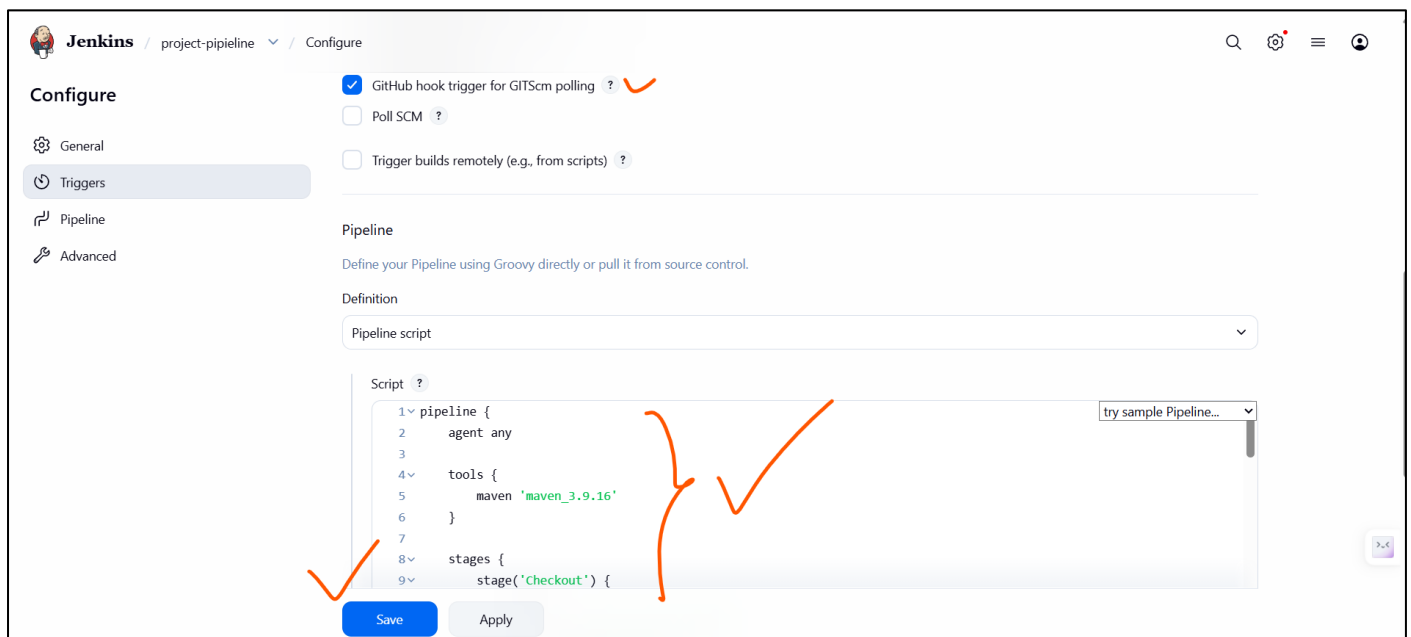
Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK

Vs code – For Jenkins CI Pipeline

C:\Users\Bhavana\Downloads\Bhavana Work\AWS Devops\AWS-Devops_MC_B17D\Kubernetes\practice\11_June_And_12_June\Jenkins-CI-project-pipeline



The image shows the Jenkins 'Configure' page for the 'project-pipeline' job. The left sidebar has 'Configure' at the top, followed by 'General', 'Triggers', 'Pipeline', and 'Advanced'. The 'Triggers' tab is selected. In the 'Triggers' section, the 'GitHub hook trigger for GITScm polling' checkbox is checked. Below this, there's a 'Pipeline' section with the instruction 'Define your Pipeline using Groovy directly or pull it from source control.' The 'Definition' dropdown is set to 'Pipeline script'. A text area for the 'Script' contains a Groovy pipeline definition. At the bottom, there are 'Save' and 'Apply' buttons.

Jenkins / project-pipeline / Configure

Configure

- General
- Triggers**
- Pipeline
- Advanced

☒ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

☐ Trigger builds remotely (e.g., from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script

```
1 pipeline {
2   agent any
3
4   tools {
5     maven 'maven_3.9.16'
6   }
7
8   stages {
9     stage('Checkout') {
```

try sample Pipeline...

Save Apply

ghbhavana-01 / AWS-Devops-Practise

Type to search

CodeIssuesPull requestsAgentsActionsProjectsWikiSecurity and qualityInsightsSettings

Okay, that hook was successfully created. We sent a ping payload to test it out! Read more about it at <https://docs.github.com/webhooks/#ping-event>.

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Models

Webhooks

Copilot

Environments

Webhooks

Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. [Learn more in our Webhooks Guide.](#)

✓ <http://13.222.226.123:8080/github-...> (push)

EditDelete

Last delivery was successful.

Jenkins / project-pipeline

Status

ChangesBuild NowConfigureDelete PipelineFull Stage ViewFavoriteOpen Blue OceanStagesRename

project-pipeline

Add description

Stage View

	Declarative: Tool Install	Checkout	Static Code Analysis - SonarQube Scan	Build Artifact	Docker Image Build	Image Scan - Trivy	Push Docker Image to DockerHub	Update Deployment Manifest
Average stage times: (full run time: ~1min 5s)	360ms	537ms	11s	10s	16s	18s	4s	784ms
#7 Jun 14 21:01 No Changes	88ms	297ms	9s	5s	14s	5s	4s	1s

project-pipeline < 7

PipelineChangesTestsArtifactsRestartSettingsHelpLogout

Branch: — 41s No changes

Commit: — a minute ago Started by user madhu

Start

Checkout

Static Code Analysis - Sonar...

Build Artifact

Docker Image Build

Image Scan - Trivy

Push Docker Image to Docker...

Update Deployment Ma...

End

Update Deployment Manifest - 1s

Restart Update Deployment Manifest

✓ > maven_3.9.16 — Use a tool from a predefined Tool Installation <1s

✓ > Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step. <1s

✓ > git config --global user.email "guntibhavana01@gmail.com" git config --global user.name "gbhavana-01" echo 'Updating deployment manifest' sed -i "s|image:.|image: dockerhub1105/practicerepo:"7"|g... — Shell Script <1s

dockerhub1105

Your personal account

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Repositories / practicerpo / General

dockerhub1105/practicerepo

Last pushed 2 minutes ago · Repository size: 1.5 GB · ☆ 0 · ↓ 137

image repo

Add a category

General

Tags

Image Management

Collaborators

Webhooks

Settings

Tags

This repository contains 10 tag(s).

Tag	OS	Type	Pulled	Pushed
7 ✓		Image	less than 1 day	2 minutes
6		Image	less than 1 day	16 minutes

DOCKER SCOUT INACTIVE

Activate

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to shared cache.

Docker Build Cloud executes builds on optimized infrastructure with dedicated per-organization resources.

Files

main

Go to file

deploymentfiles

deploy.yaml ✓

service.yaml

src

target

Dockerfile

Jenkinsfile

Jenkinsfile_Argocd_project

Jenkinsfile_docker_session

Jenkinsfile_env_vars

README.md

pom.xml

Code

Blame

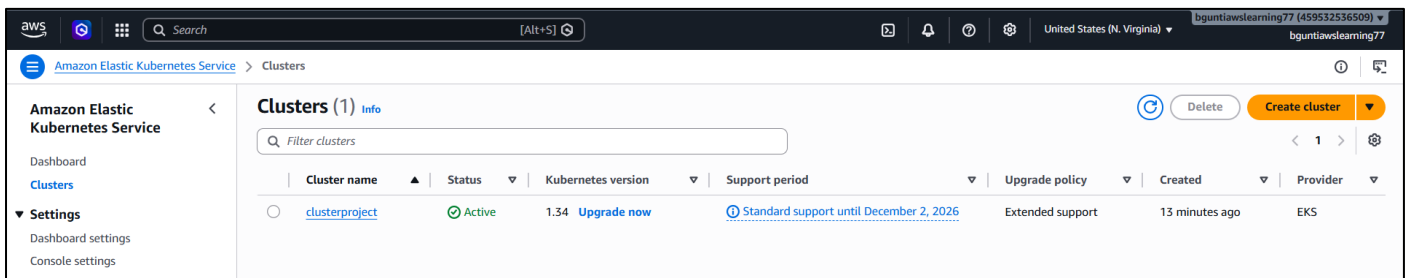
29 lines (26 loc) · 452 Bytes

```
1 ---
2
3 apiVersion: apps/v1
4 kind: Deployment
5 metadata:
6   name: fb-deploy
7   labels:
8     app: fb-app
9     env: prod
10 spec:
11   replicas: 3
12
13   selector:
14     matchLabels:
15       app: fb-app
16       env: prod
17   template:
18     metadata:
19       labels:
20         app: fb-app
21         env: prod
22         version: v2_vc
23
24   spec:
25     containers:
26     - name: facebook-container
27       image: dockerhub1105/practicerepo:'7' ✓
28       ports:
29       - containerPort: 8080
```

Create Jenkins CD Pipeline

12.Create EKS cluster

```
[root@utilserver ~]# eksctl create cluster --name clusterproject --nodegroup-name b17dng --node-type m7i-flex.large --nodes 4 --managed
```



```
[root@utilserver ~]# eksctl get cluster
```

```
NAME          REGION    EKSTCL CREATED
```

```
clusterproject us-east-1  True
```

```
[root@utilserver ~]# kubectl get nodes
```

```
NAME                                STATUS  ROLES  AGE  VERSION
```

```
ip-192-168-24-83.ec2.internal  Ready  <none>  4m25s  v1.34.8-eks-3385e9b
```

```
ip-192-168-27-125.ec2.internal  Ready  <none>  4m26s  v1.34.8-eks-3385e9b
```

```
ip-192-168-53-90.ec2.internal  Ready  <none>  4m24s  v1.34.8-eks-3385e9b
```

```
ip-192-168-63-80.ec2.internal  Ready  <none>  4m25s  v1.34.8-eks-3385e9b
```

```
[root@utilserver ~]# kubectl create namespace argocd
```

```
namespace/argocd created
```

```
[root@utilserver ~]# kubectl get ns
```

```
NAME          STATUS  AGE
```

```
argocd        Active  18s
```

```
default       Active  10m
```

```
kube-node-lease  Active  10m
```

```
kube-public     Active  10m
```

```
kube-system     Active  10m
```

```
[root@utilserver ~]# kubectl apply -f https://raw.githubusercontent.com/argoproj/argo-cd/stable/manifests/install.yaml -n argocd
```

```
[root@utilserver ~]# kubectl get all -n argocd
```



```
[root@utilserver ~]# kubectl get svc -n argocd
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
argocd-applicationset-controller	ClusterIP	10.100.186.197	<none>	7000/TCP,8080/TCP	2m15s
argocd-dex-server	ClusterIP	10.100.242.207	<none>	5556/TCP,5557/TCP,5558/TCP	2m15s
argocd-metrics	ClusterIP	10.100.157.33	<none>	8082/TCP	2m15s
argocd-notifications-controller-metrics	ClusterIP	10.100.199.19	<none>	9001/TCP	2m15s
argocd-redis	ClusterIP	10.100.246.173	<none>	6379/TCP	2m15s
argocd-repo-server	ClusterIP	10.100.242.52	<none>	8081/TCP,8084/TCP	2m15s
argocd-server	ClusterIP	10.100.118.37	<none>	80/TCP,443/TCP	2m15s
argocd-server-metrics	ClusterIP	10.100.237.17	<none>	8083/TCP	2m15s

```
[root@utilserver ~]# kubectl patch svc argocd-server -n argocd -p '{"spec": {"type": "LoadBalancer"}}'
```

```
service/argocd-server patched
```

```
[root@utilserver ~]# kubectl get svc -n argocd
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
argocd-applicationset-controller	ClusterIP	10.100.186.197	<none>	7000/TCP,8080/TCP	3m4s
argocd-dex-server	ClusterIP	10.100.242.207	<none>	5556/TCP,5557/TCP,5558/TCP	3m4s
argocd-metrics	ClusterIP	10.100.157.33	<none>	8082/TCP	3m4s
argocd-notifications-controller-metrics	ClusterIP	10.100.199.19	<none>	9001/TCP	3m4s
argocd-redis	ClusterIP	10.100.246.173	<none>	6379/TCP	3m4s

argocd-repo-server ClusterIP 10.100.242.52 <none>
8081/TCP,8084/TCP 3m4s

argocd-server LoadBalancer 10.100.118.37
a98d5f6c333984dc1aba82d175724314-1758031739.us-east-1.elb.amazonaws.com
80:31841/TCP,443:30100/TCP 3m4s

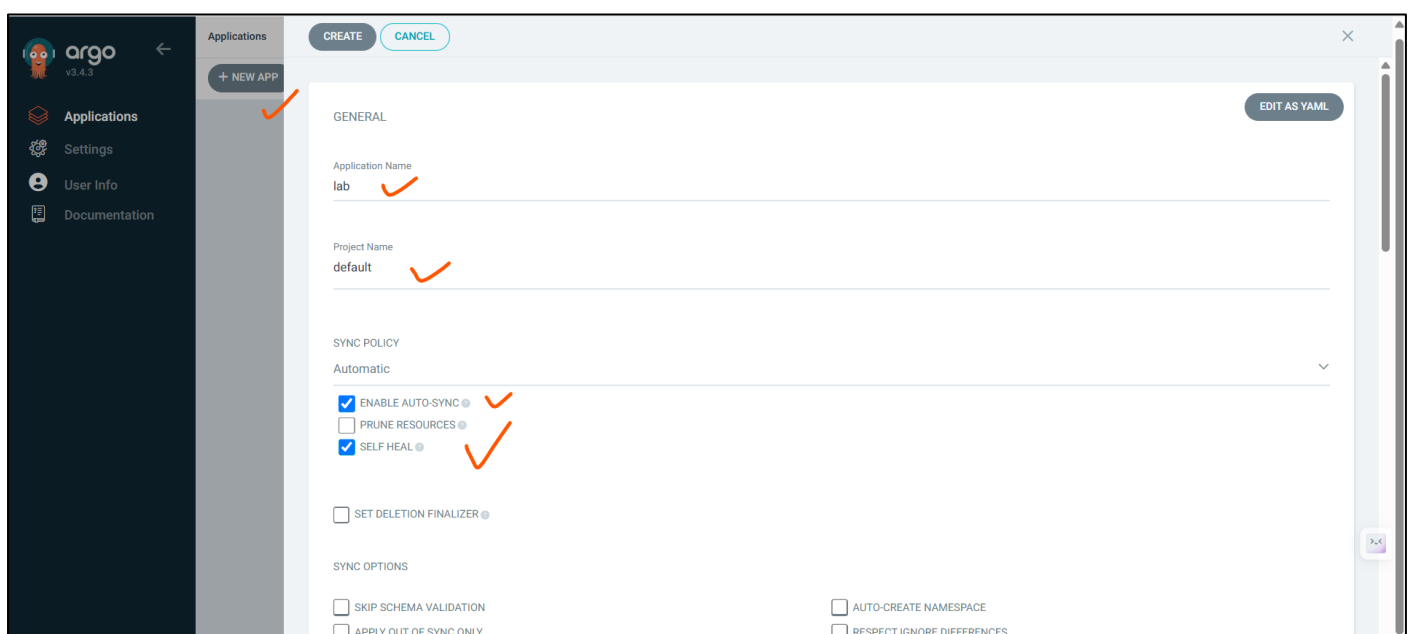
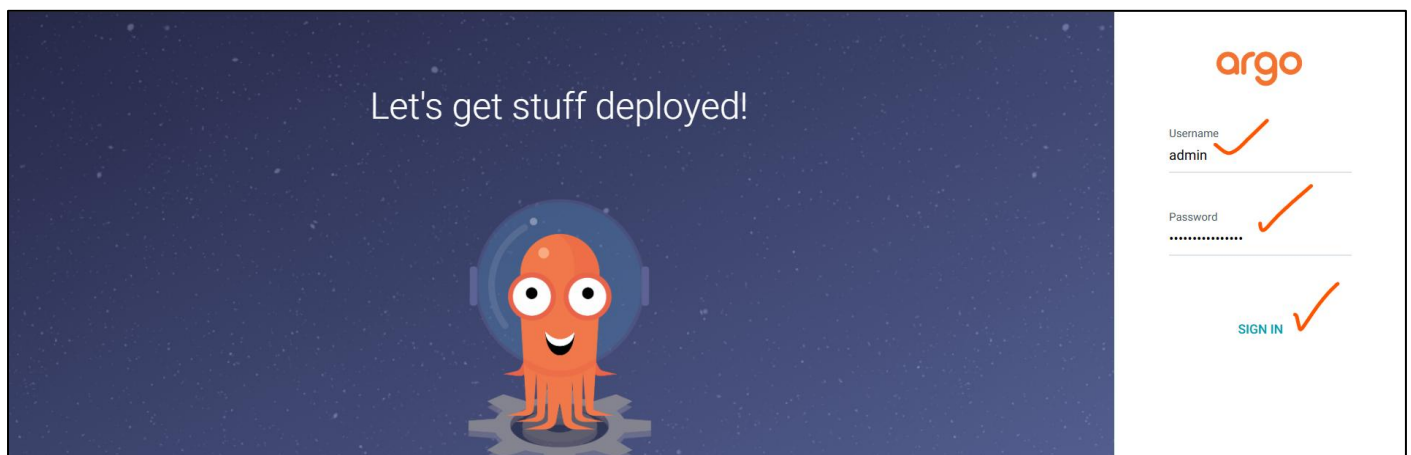
argocd-server-metrics ClusterIP 10.100.237.17 <none>
8083/TCP 3m4s

```
[root@utilserver ~]# kubectl -n argocd get secret argocd-initial-admin-secret -o  
jsonpath="{.data.password}" | base64 -d; echo
```

gdQZwo6wMrK5eI5e

Argo Cd loadbalancer endpoint

a98d5f6c333984dc1aba82d175724314-1758031739.us-east-1.elb.amazonaws.com



argo v3.4.3

Applications

+ NEW APP

CREATE CANCEL

SOURCE

Repository URL
https://github.com/gbhavana-01/mindcircuit17d.git ✓

Revision
HEAD ✓

Path
deploymentfiles ✓

DESTINATION

Cluster URL
https://kubernetes.default.svc ✓

Namespace
lab ✓

Applications

+ NEW APP SYNC APPS REFRESH APPS Search applications...

lab

Project: default

Labels:

Status: Missing OutOfSync Syncing

Repository: https://github.com/gbhavana-01/mindcircuit17...

Target Revision: HEAD

Path: deploymentfiles

Destination: in-cluster

Namespace: lab

Created At: 06/14/2026 22:54:32 (a few seconds ago)

Last Sync: 06/14/2026 22:54:34 (in a few seconds)

SYNC REFRESH DELETE

Applications / Q lab

DETAILS DIFF SYNC SYNC STATUS HISTORY AND ROLLBACK DELETE REFRESH

APP HEALTH Missing

SYNC STATUS OutOfSync from HEAD (01e075f)

LAST SYNC Syncing

Auto sync is enabled.

Author: gbhavana-01 <gunthbhavana01@gmail.com> -

Comment: Updated deployment manifest with build number 7

Running a few seconds ago (Sun Jun 14 2026 22:54:48 GMT+0530)

one or more objects failed to apply, reason: namespaces "lab" not found. Retrying attempt #2 at 5:24PM.

lab

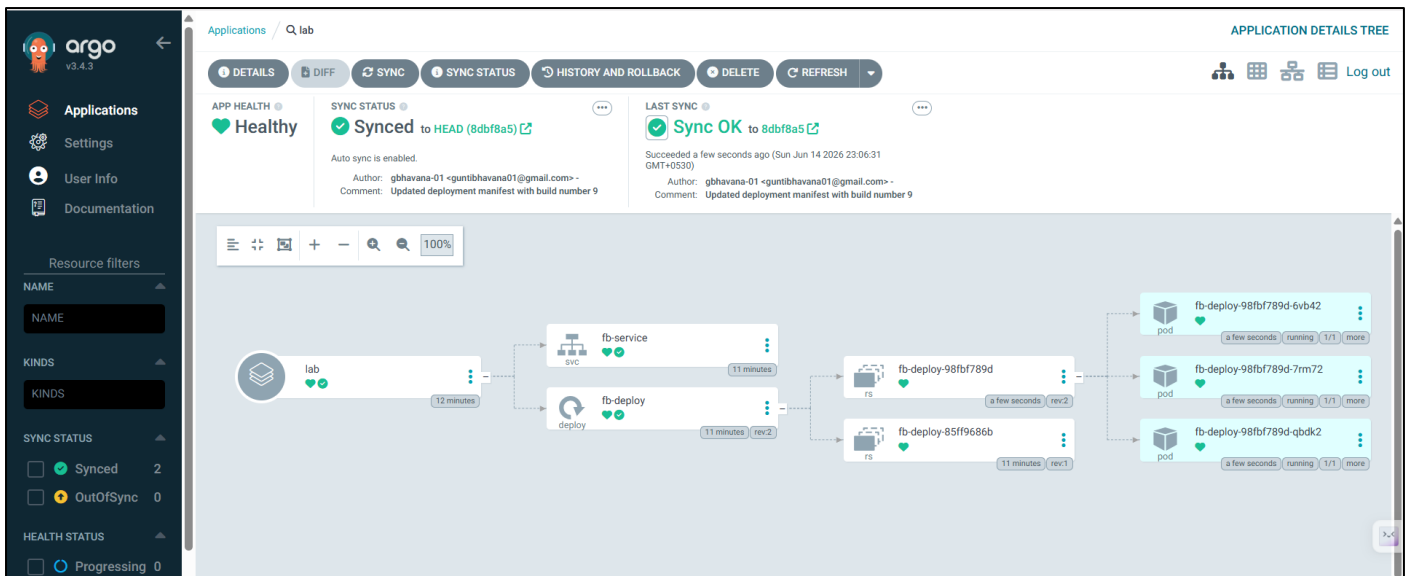
fb-service

fb-deploy

a few seconds

[root@utilserver ~]# kubectl create namespace lab

namespace/lab created



```
[root@utilserver ~]# kubectl get svc -n lab
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)
fb-service	LoadBalancer	10.100.231.5	a1ba21f9798f04acc84f6e849bd9cca7-644309754.us-east-1.elb.amazonaws.com	80:30641/TCP

```
fb-service LoadBalancer 10.100.231.5 a1ba21f9798f04acc84f6e849bd9cca7-644309754.us-east-1.elb.amazonaws.com 80:30641/TCP 23s
```

```
[root@utilserver ~]# kubectl get pods -n lab
```

NAME	READY	STATUS	RESTARTS	AGE
fb-deploy-8df787dcb-44klw	1/1	Running	0	64s
fb-deploy-8df787dcb-g8kqp	1/1	Running	0	64s
fb-deploy-8df787dcb-zdt9g	1/1	Running	0	64s

```
[root@utilserver ~]# kubectl get svc -n lab
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)
fb-service	LoadBalancer	10.100.231.5	a1ba21f9798f04acc84f6e849bd9cca7-644309754.us-east-1.elb.amazonaws.com	80:30641/TCP

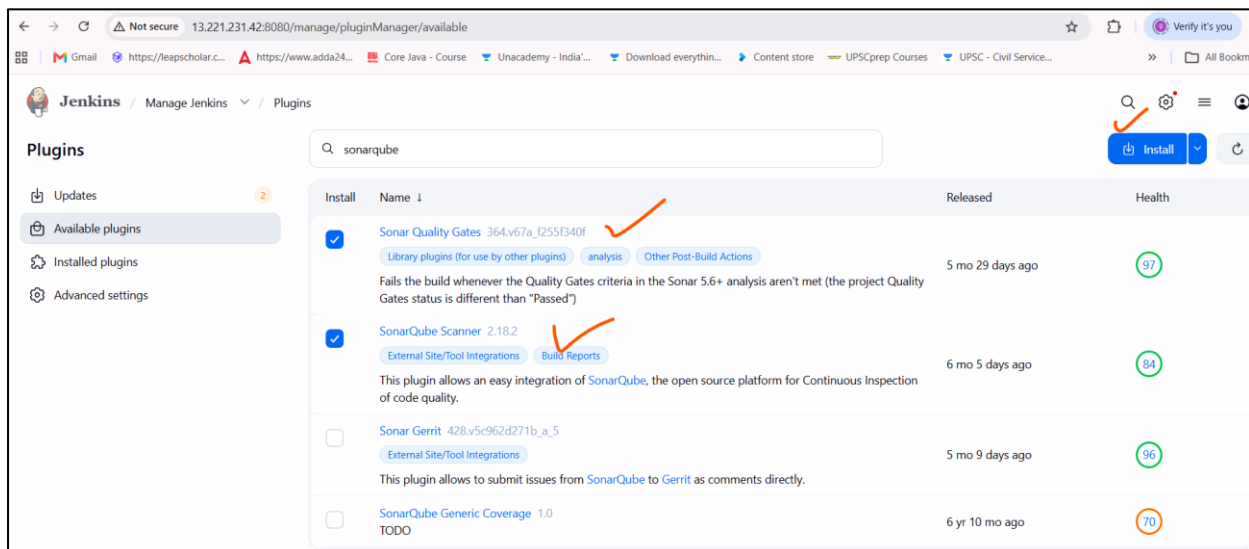
```
fb-service LoadBalancer 10.100.231.5 a1ba21f9798f04acc84f6e849bd9cca7-644309754.us-east-1.elb.amazonaws.com 80:30641/TCP 73s
```

To See the application in lab ns – pods – application – loadbalancer endpoint
a1ba21f9798f04acc84f6e849bd9cca7-644309754.us-east-1.elb.amazonaws.com

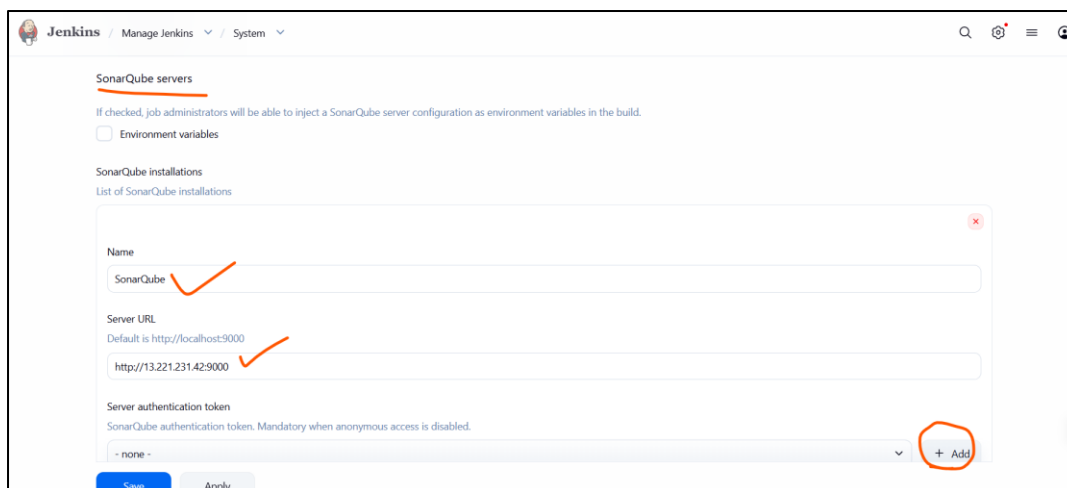
MY NAME IS Madhu Kiran , A PASSIONATE DEVOPS Trainer and LEARNER

Jenkins CI pipeline Improvement

- In jenkins – install plugin - SonarQube Scanner for jenkins, Sonar Quality Gates plugin



- Add SonarQube in Manage Jenkins



Jenkins / Manage Jenkins / System

☐ Environment variables

SonarQube installations

List of SonarQube installations

Name
SonarQube

Server URL
Default is http://localhost:9000
http://13.221.231.42:9000

Server authentication token
SonarQube authentication token. Mandatory when anonymous access is disabled.
- none -

Advanced

Save Apply

Add Credentials

Select a type of credential

- Username with password
Commonly used for authentication to services like Git, APIs, or registries.
- GitHub App
- OpenShift Username and Password
- SSH Username with private key
- Secret file
- Secret text**
- X.509 Client Certificate

Next

← Add Secret text ×

Scope ?
Global (Jenkins, nodes, items, all child items, etc)

Secret
.....

ID ?
labsonartoken

Description ?
labsonartoken

Create

Jenkins / Manage Jenkins / System

☐ Environment variables

SonarQube installations

List of SonarQube installations

Name
SonarQube

Server URL
Default is http://localhost:9000
http://13.221.231.42:9000

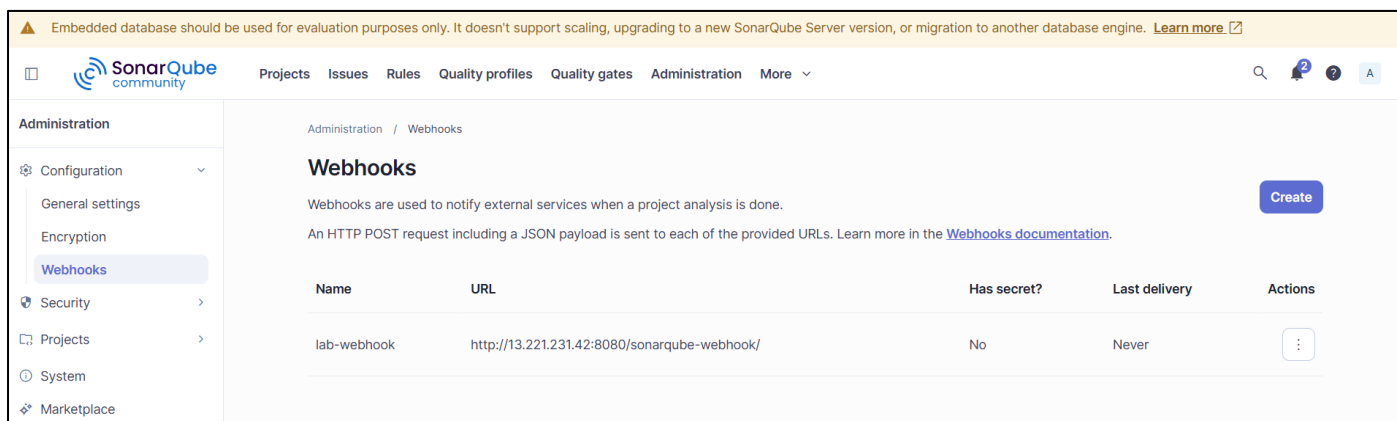
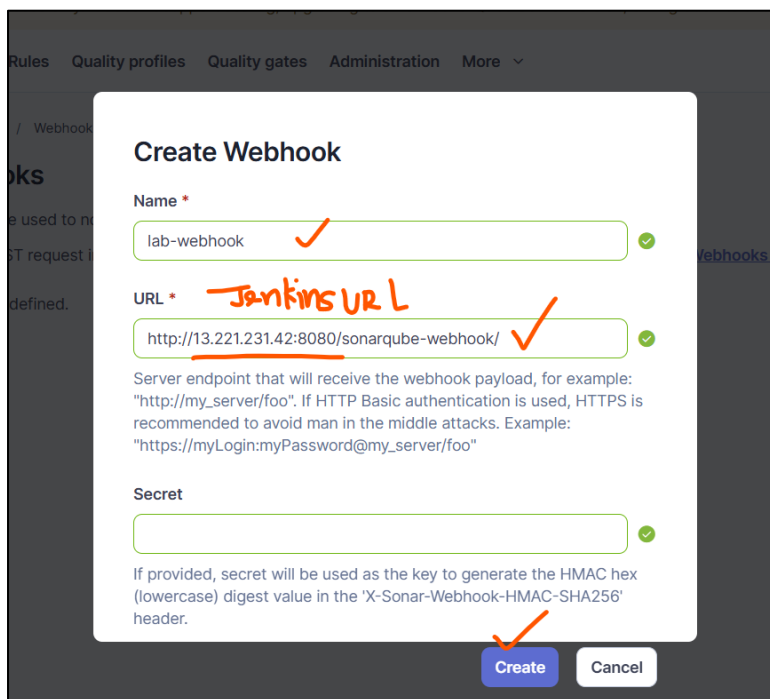
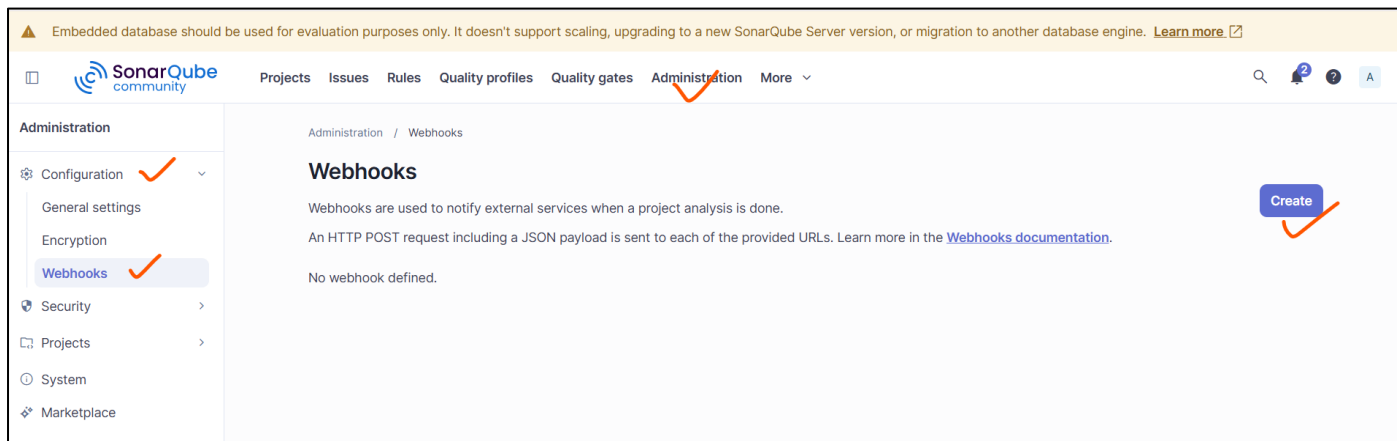
Server authentication token
SonarQube authentication token. Mandatory when anonymous access is disabled.
labsonartoken

Advanced

Save Apply

```
[root@utilserver ~]# systemctl restart jenkins
```

SonarQube -- Configure Sonar webhook to tell Jenkins Quality Gate is passed



tice > 11_June_And_12_June > Jenkins-CL-improved

```
pipeline {
  stages {
    stage('Checkout') {
    }

    stage('Build & Unit Test') {
      steps {
        sh 'mvn clean test'
      }
    }

    stage('SonarQube Scan') {
      steps {
        withSonarQubeEnv('SonarQube') { // sonarQube server name defined in Jenkins global tool configuration
          sh '''
            mvn clean sonar:sonar
          '''
        }
      }
    }

    stage('Quality Gate') {
      steps {
        timeout(time: 5, unit: 'MINUTES') {
          waitForQualityGate abortPipeline: true
        }
      }
    }

    stage('Build the artifact') {
      steps {
        echo 'Build the artifact'
      }
    }
  }
}
```

```
stage('Docker Image Build') {
  steps {
    echo 'Build the docker image'
    sh 'docker build -t dockerhub1105/practicerepo: ${BUILD_NUMBER} -f Dockerfile .'
  }
}

stage('Image Scan - Trivy') {
  steps {
    echo 'Scan the docker image - Trivy'
    sh 'trivy image --severity, HIGH dockerhub1105/practicerepo:${BUILD_NUMBER}'
  }
}
```


Jenkins / project-pipeline / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Definition

Pipeline script

Script ?

```
1 pipeline {
2   agent any
3
4   tools {
5     maven 'maven_3.9.16'
6   }
7
8   stages {
9     stage('Checkout') {
10      steps {
11        echo 'checkout code from git'
12        git branch: 'main',
13            url: 'https://github.com/gbhavana-01/mindcircuit17d.git'
14      }
15    }
16  }
17}
```

Use Groovy Sandbox ?

Save Apply

New/Improved Jenkins CI

Jenkins / project-pipeline

Status

project-pipeline

Add description

Stage View

Average stage times: (full run time: ~58s)

	Declarative: Tool Install	Checkout	Build & Unit Test	SonarQube Scan	Quality Gate	Build Artifact	Docker Image Build	Image Scan - Trivy	Push Docker Image to DockerHub	Update Git Repo Deployment File
#12 Jun 15 12:20 1 commit	262ms	657ms	3s	11s	330ms	4s	13s	9s	4s	1s
#11 Jun 15 12:18 No Changes										

SonarQube Quality Gate

facebookapp Passed

server-side processing: Success

Files

main

Go to file

- deploymentfiles
 - deploy.yaml
 - service.yaml
- src
- target
 - Dockerfile
 - Jenkinsfile
 - Jenkinsfile_Argocd_project
 - Jenkinsfile_docker_session
 - Jenkinsfile_env_vars
 - README.md
 - pom.xml

Code Blame 29 lines (26 loc) · 451 Bytes

```
1
2 ---
3 apiVersion: apps/v1
4 kind: Deployment
5 metadata:
6   name: fb-deploy
7   labels:
8     app: fb-app
9     env: prod
10 spec:
11   replicas: 3
12
13   selector:
14     matchLabels:
15       app: fb-app
16       env: prod
17   template:
18     metadata:
19       labels:
20         app: fb-app
21         env: prod
22         version: v2_vc
23
24   spec:
25     containers:
26     - name: facebook-container
27       image: dockerhub1105/practicerepo:12
28       ports:
29       - containerPort: 8080
```

argo
v3.4.3

Applications

Settings

User Info

Documentation

Resource filters

NAME

NAME

KINDS

KINDS

SYNC STATUS

Applications / Q lab

APPLICATION DETAILS TREE

Log out

DETAILS

DIFF

SYNC

SYNC STATUS

HISTORY AND ROLLBACK

DELETE

REFRESH

APP HEALTH

Healthy

SYNC STATUS

Synced to HEAD (9ad8184)

Auto sync is enabled

Author: gbhavana-01 <guntilbhavana01@gmail.com>

Comment: Updated deployment file with build number 12

LAST SYNC

Sync OK to 9ad8184

Succeeded 3 minutes ago (Mon Jun 15 2026 12:22:32 GMT+0530)

Author: gbhavana-01 <guntilbhavana01@gmail.com>

Comment: Updated deployment file with build number 12

lab

an hour

fb-service

SVC

an hour

fb-deploy

FS

an hour

fb-deploy-64cdb49dd6

FS

3 minutes

rev2

fb-deploy-8df787dcb

FS

an hour

rev1

fb-deploy-64cdb49dd6-k5lh2

pod

3 minutes

running

1/1

more

fb-deploy-64cdb49dd6-nc177

pod

3 minutes

running

1/1

more

fb-deploy-64cdb49dd6-rqt6v

pod

3 minutes

running

1/1

more

←

→

↺

⚠ Not secure

a1ba21f9798f04acc84f6e849bd9cca7-644309754.us-east-1.elb.amazonaws.com

☆

🔖

Verify it's you

⋮

📧 Gmail

🌐 https://leapscholar.c...

🔴 https://www.adda24...

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